Bayesian Knowledge Tracing

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Байесовское Отслеживание Знаний

13я Ежегодная Летняя Школа LearnLab, Питтсбург, ПА 2017

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Летняя Школа PSLC 2006

A little bit of context (1)

- Tutoring session
  - Human tutor or an Intelligent Tutoring System
- Student interacts with a problem
  - Single-step or multi-step
- Tutor tracks student's progress
- Tutor provides feedback
- Tutor gives advice

A little bit of context (2)

- Intelligent Tutoring System
  - Model tracing – interpret student behavior by comparing to the ***model of the problem***
  - Knowledge Tracing – infer student knowledge based on student performance; knowledge components (skills) associated with problem (steps)
- Cognitive Mastery
  - Deliver problems that address skills not mastered yet
  - Skill is considered mastered if inferred probabilistic value ≥ threshold

Agenda

- **Bayesian Knowledge Tracing (BKT)**
- BKT Variants
- Fitting BKT at Scale (`hmm-scalable`)
- Advanced Topics

Bayesian Knowledge Tracing

- **Goal**: for every skill the tutored problem (step) is associated with
  - Does the student know the skill? (mastery)
  - Will the student apply the skill correctly? (performance)
- Given the sequence of opportunities below (0-incorrect, 1-correct), answer the 2 questions above

1 0 1 0 1 1

Bayesian Knowledge Tracing
Definition

Bayesian Knowledge Tracing Model $\subseteq$ Hidden Markov Model $\subseteq$ Bayes Net $\subseteq$ Graphical Model

When modeling skill acquisition

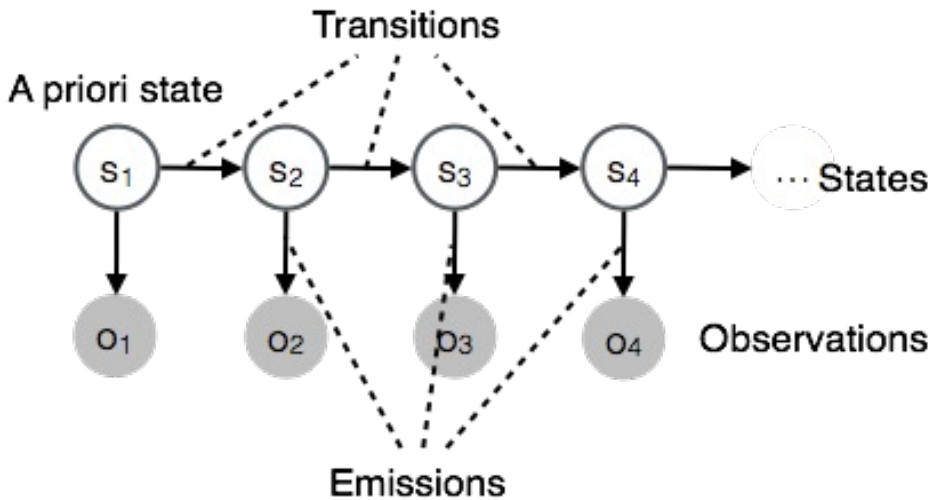
- One binary latent (unobserved) variable represents mastery (skill is ***known***|***unknown***), N=2
- One observed binary variable represents student performance (skill is applied ***correctly***|***incorrectly***), M=2

Data

- Time-series data of students applying skills to solve problems
  - Multiple time-series per skill from different students

Bayesian Knowledge Tracing
Representation

Unrolled view of BKT



Priors – Π

Known	Unknown
pL_0	1- pL_0

Transitions – A

	to Known	to Unknown
from Known	1-pF	pF
from Unknown	**pT**	1-pT

Emissions – B

	Correct	Incorrect
Known	1-pS	**pS**
Unknown	**pG**	1-pG

- 4 parameters
  - pL_o, pInit
  - pT, pLearn
  - pF, pForget
  - pS, pSlip
  - pG, pGuess
- Values ∈[0,1]
- Rows sum to 1
- Forgetting (pF) = 0

Bayesian Knowledge Tracing
Updating masteries, predicting

Parameters

Π	A		B	
pL_0	$\neg pF$	pF	$\neg pS$	pS
$\neg pL_0$	pT	$\neg pT$	pG	$\neg pG$

Outputs

L	C
pL	pC
$\neg pL$	$\neg pC$

Helpers

Le	Be(corr)	Be (incorr)
$pL\|e$	$\neg pS$	pS
$\neg pL\|e$	pG	$\neg pG$

	Matrix Version	**Simplified Computation**
1. Initialize	$L=\Pi$	$pL=pL_0$
2. Correct		
a. Predict	$C=L^T\times B$	$pC=pL\cdot(1-pS)+(1-pL)\cdot pG$
b. Pick Be	$Be=B[,1]$	
c. Compute Le	$Le=(L.\times Be)./(L^T\times Be)$	$p(L\|e=correct)=pL\cdot(1-pS)/[\,pL\cdot(1-pS)+(1-pL)\cdot pG\,]$
d. Update L	$L=Le\times A$	$pL=p(L\|e)\cdot(1-pF)+(1-p(L\|e))\cdot pT$
3. Incorrect (a and d same as in 2)		
b. Pick Be	$Be=B[,2]$	
c. Compute Le		$p(L\|e=incorrect)=pL\cdot pS/[\,pL\cdot pS+(1-pL)\cdot(1-pG)\,]$
4. Unknown (a-c same as in 2)		
d. Update L	$L=A^T\times L$	$pL=pL\cdot(1-pF)+(1-pL)\cdot pT$
5. Unknown/guess		
b. Pick Be	$Be=B[,argmax(C)]$	
continue as 2 or 3, depending on argmax(C)		

Legend

× – vector/matrix multiply
.× – multiply element by element
./ – divide element by element
Xe – X given evidence
L^T – transpose
X[,i] – i^{th} column of X

Bayesian Knowledge Tracing

http://yudelson.info/quickbkt.html

This is a quick helper interface for tracing/debugging/developing Bayesian Knowledge Tracing models [1]. It covers computing of the skill mastery (latent), performance (observation), as well as forward and backward variables [2].

Computations are done with the help of Numeric Javascript

pInit 0.5
pLearn 0.4
pForget 0.0
pSlip 0.1
pGuess 0.2

Parameters

Observations 1 0 1 0 1 1

Observations

0's and 1's separated by spaces

Compute | Update parameters. Expectation Maximization (Beta)

Likelihood of the data given the model: 0.006213451680

#	Right/ wrong	pKnown	pCorrect	Forward Variables		Backward Variables		Scale c_t	Fwd. Variables Scaled		Bwd. Variables Scaled	
				$\alpha_{t,i=1}$	$\alpha_{t,i=2}$	$\beta_{t,i=1}$	$\beta_{t,i=2}$		$\hat{\alpha}_{t,i=1}$	$\hat{\alpha}_{t,i=2}$	$\hat{\beta}_{t,i=1}$	$\hat{\beta}_{t,i=2}$
0†	n/a	0.500000	0.550000	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	
1	1	0.890909	0.823636	0.450000	0.100000	0.007290	0.029330	1.818182	0.818182	0.181818	0.015670	0.063044
2	0	0.703093	0.692165	0.049000	0.048000	0.072900	0.055028	5.670103	0.505155	0.494845	0.137935	0.104120
3	1	0.948525	0.863968	0.061380	0.005760	0.081000	0.215568	1.444742	0.914209	0.085791	0.134910	0.359040
4	0	0.818368	0.772858	0.006368	0.002765	0.810000	0.381600	7.351202	0.697280	0.302720	1.187555	0.559470
5	1	0.971798	0.880259	0.006727	0.000332	0.900000	0.480000	1.293899	0.952997	0.047003	1.161507	0.619470
6	1	0.996155	0.897309	0.006174	0.000040	1.000000	1.000000	1.136029	0.993592	0.006408	1.136029	1.136029

Bayesian Knowledge Tracing. Special Note on Predicting

- Updating L (pL) when the observation is known
  - In the tutor, the observation (correct/incorrect) is available for the pL update
    - Given 1..n skill attempts, predict n+1st
  - When comparing models making the observation available puts BKT at an unfair advantage
    - Especially, when comparing to logistic regression-based AFM
    - Less unfair when comparing to PFA (two learn rates)
  - Rule 4 from previous slide is too harsh
    - Matrix B (emissions) is ignored
    - Accuracy takes a considerable hit
  - Rule 5 is a compromise
    - Accuracy is better than under Rule 4 (but worse than under Rules 2&3)

Agenda

- Bayesian Knowledge Tracing (BKT)
- **BKT Variants**
- Fitting BKT at Scale (`hmm-scalable`)
- Advanced Topics

BKT Variants

- Standard BKT: one BKT model per skill
- Individualized BKT: parameterize students
  - Multiplex pInit (Pardos & Heffernan, 2010)
  - Fit BKT parameters within not across students (Lee & Brunskill 2012)
  - Fit per-skill and per-student parameters (Corbett & Anderson 1995; Yudelson et al. 2013)
- Accounting for item effects
  - Introduce an item node that affects performance (Pardos & Heffernan 2011)
- Dealing with noise in the data
  - From unigram observations to n-gram observations (Falakmasir et al. 2015)

Individualized BKT (1)

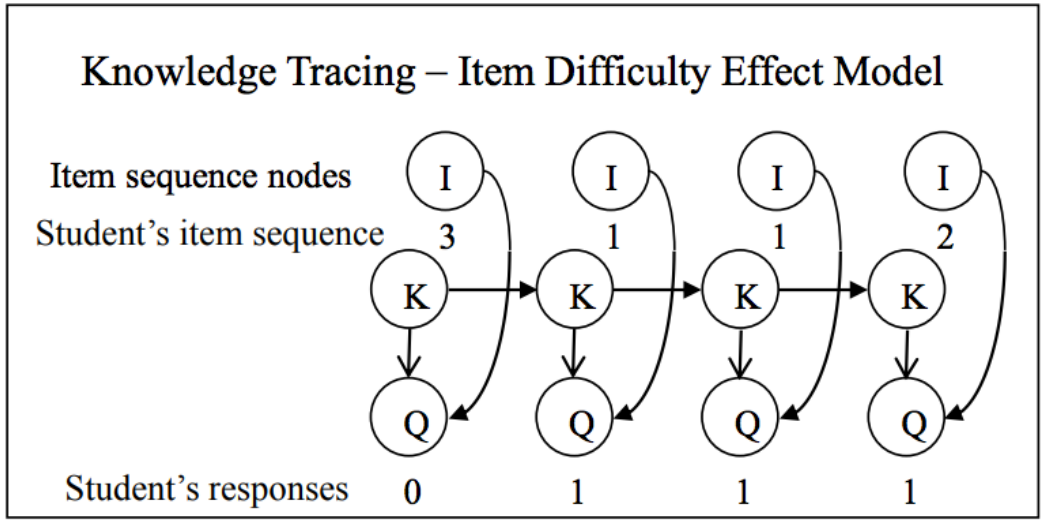
- Individualization – accounting for student-specific variation in performance
- Mulitplex pInit (Pardos & Heffernan, 2010)
  - Students are classified into low-pInit and high-pInit using first few attempts (often the first)
  - pKnown recomputed if classification changes from the default
- Fit BKT parameters within students (Lee & Brunskill 2012)
  - Hard to operationalize (chronic cold start problem)
  - Theoretical upper boundary
  - Resulting benefit of a more optimal problem selection is significant

Individualized BKT (2)

- Per-student and per-skill parameters
  - Standard 4 per-skill BKT parameters remain
  - Additional 4 per-student parameters
  - Pairing function combines the two
    - Corbett & Anderson 1995 – in odds space
    - Yudelson et al 2013 – in log-odds space
  - Fitting procedure.
    - Corbett & Anderson 1995 – fit once
    - Yudelson et al 2013 – block coordinate descent
  - Improvements are significant
  - Operationalization is tricky
    - Corbett & Anderson 1995 – log(student parameter)~raw error rate
    - Yudelson et al 2013 – fitting procedure pulls too much variance away from skill parameters
- Skill or Student, a limited test (Yudelson, 2016)
  - Parameter ~ $B_0 + B_1 \cdot \text{Skill} + B_2 \cdot \text{Student} + B_3 \cdot \text{Skill} * \text{Student}$
  - B_0 and B_3 are small (good)
  - Student weight B_2 is twice as high

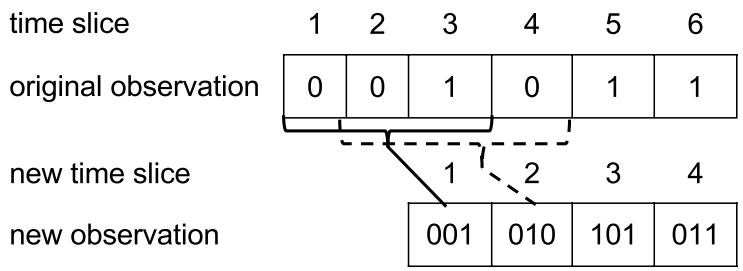
Accounting for item effects

- Introduce an item node that affects performance (Pardos & Heffernan 2011)
  - More complex HMM topology
  - Tested on ASSISTments and Cognitive Tutor data
  - Trained using BNT toolkit for Matlab/Octave



Dealing With Noise in the Data

- From unigram to n-gram observations: Feature compensation (often used in NL processing)



- >2 states and >2 observations

	to known	to unknown
from known	1	0
from unknown	0.4	0.6

	to 1	to 2	to 3	to 4
from 1	1	0	0	0
from 2	>0	>0	>0	0
from 3	>0	>0	>0	0
from 4	0	>0	>0	>0

- Best C-V'd accuracy achieved on KDD 2010 dataset

Model	Par/skill	C-V strat.	AIC	BIC	Acc.	RMSE
BKT	4	item	15380089	15478083	0.8609	0.3293
Spectral BKT	37	item	15853433	16758456	0.9208	0.2472
BKT	4	student	13947080	14045074	0.8659	0.3153
Spectral BKT	37	student	11553442	12458465	0.9196	0.2405

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- **Fitting BKT at Scale (`hmm-scalable`)**
- Advanced Topics

`hmm-scalable`

- http://yudelson.info/hmm-scalable
- Toolkit for building Bayesian Knowledge Tracing (BKT) models using Big Datasets
- Extensively tested on educational data
- How big is a Big Dataset?
  - KDD Cup 2010 data. Challenge Set A/B – 9M/20M records 3K/6K students
  - 2010 Carnegie Learning Cognitive tutor data
    92.5 M records, 74 000 students (unfiltered)
  - Synthetic dataset – 902M records, 100K students
- How fast can `hmm-scalable` fit a model?
  - KDD Cup – 1 minute
  - 2010 Carnegie Learning unfiltered – 5 minutes (multi-core mode)
  - Synthetic data – 20 minutes (multi-core mode)

Obtaining and compiling `hmm-scalable`

Get the source code from IEDMS GitHub repository

```
$> git clone https://github.com/myudelson/hmm-scalable
```

OR download and unzip

Alternatively download an executable and readme from DataShop

External Tools > Fitting Bayesian Knowledge Tracing models at scale

`http://pslcdatashop.web.cmu.edu/ExternalTools?toolId=1`

Compile

```
$> make all
```

Operating system notes

- Linux – `gcc` with `Open MP` should be there by default
- Mac OS X – refer to hpc.sourceforge.net
- Windows – install `cygwin` with `awk`, **`make`**, `wc`, `time`, `unzip` commands

Getting KDD Cup 2010 Data

KDD Cup 2010 on hosted by LearnLab's DataShop [pslcdatashop.web.cmu.edu/KDDCup] Largest freely available dataset of learner data donated by Carnegie Learning, Inc.

- Download Bridge to Algebra challenge set bridge_to_algebra_2008_2009.zip (461Mb)
- ~20 000 000 rows, ~6 000 students

Data is placed in subfolder `d`, models will be in subfolder `m`, prediction files in subfolder `p`, code is in the root that contains these subfolders

Subset KDD Cup 2010 Data

Instead of unzipping a 5.3Gb training file, run:

```
$> time unzip -q -c d/bridge_to_algebra_2008_2009.zip
bridge_to_algebra_2008_2009_train.txt | awk "-F\t"
'BEGIN{OFS="\t"} {if(NR==1)next; skill=$20; gsub("~~",
"~", skill); skill=(skill==""")?".":($3"__"skill); print 2-
$14,$2,$3"__"$4,skill;}' > d/b89_train.txt
```

Reduces rich data to tab-delimited SSSS: success, student, step, skill(s). Multiple skills are ~-delimited. No skill label is denoted by . (dot).

Resulting text file is 2.9Gb.

Note: skills are prefixed with unit and section name. Identical skill labels could mean different things in different sections.

Training using `hmm-scalable`

Let's start from EM solver

```
$> ./trainhmm -s 1.1 -d ~ -m 1 -p 1 d/b89_train.txt
m/b89_modelEM.txt p/b89_predictEM.txt
```

Here, `-s 1.1` is an EM solver, `-d ~` is setting skill delimiter, `-m 1` asks for fit metrics, and `-p 1` requires predictions to be written into a file

It takes a little under 2.5 minutes, fitting alone 1 minute

```
trained model LL=6903034.9861273 (4490671.7596719),
AIC=13820821.972255, BIC=13930074.307203, RMSE=0.318739
(0.352797), Acc=0.870607 (s)
timing: overall 139.576931 sec, read 42.205482 sec, fit 60.997095
sec, predict 36.144902 sec
```

Here, LL – sum of negative log-likelihoods, AIC and BIT are Akaike and Bayesian Information Criterions, RMSE - root mean squared error, Acc – accuracy.

In braces are the RMSE and Accuracy for the rows without skill labels. Otherwise the metric is given for all rows, even if skill label was omitted.

Using Different "Solvers" (1)

- Baum-Welch (EM) `-s 1.1` – classical algorithm
- Stochastic Gradient Descent `-s 1.2`
  - Scales gradients so that the maximum step of 1 does not take any of them over 1 or below 0 (finds n to divide all parameters by 10^n)
  - Halves the step until either Wolfe Conditions are met (a step is far enough but not too far) or 20 attempts. If only first WC is met, us that (Nocedal & Wright, 1999).
  - Re-align parameters so that rows sum to 1 (project onto simplex)
  - Attempts to make large leaps in parameter space (Wolfe Conditions counter-balance that)
- Conjugate Gradient Descent `-s 1.3.[1-4]`
  - Make first step as in Stochastic Gradient Descent
  - Computes step direction on subsequent next step using formulas
    - Fletcher-Reeves `-s 1.3.1`
    - Polak-Ribière `-s 1.3.2`
    - Hestenes-Stiefel `-s 1.3.3`
    - Dai-Yuan `-s 1.3.4`
  - Makes large leaps in parameter space
- Gradient Descent using Lagrangian multipliers `-s 1.4`
  - Uses gradients in a manner that does not need realignment of parameters (Levinson et al., 1983)
  - Smooth[er] parameter changes, just like EM
- Temporarily excluded Barzilai-Borwein and exponentiated Gradient updates

Using Different "Solvers" (2)

Now let's try a Stochastic Gradient Descent solver

```
$> ./trainhmm -s 1.2 -d ~ -m 1 -p 1 d/b89_train.txt m/b89_modelGD.txt
p/b89_predictGD.txt
```

It takes a little longer

```
trained model LL=7230246.8188073 (4817883.5923708), AIC=14475245.637615,
BIC=14584497.972563, RMSE=0.328491 (0.368380), Acc=0.863745 (0.818622)
timing: overall 169.495414 sec, read 44.967394 sec, fit 100.809863 sec,
predict 23.490269 sec
```

The overall accuracy is a little worse 0.863745 (vs. 0.870607), just as the overall RMSE – 0.328491 (vs. 0.318739).

- How much is ***a little*** and what does it mean
  - Individualized BKT (Yudelson et al., AIED 2013)
    - Increment in the number of rows we are predicting correctly (~50K)
    - **When** the changes happen: earlier attempts – more valuable
  - Small Improvement for the Model Accuracy Big Difference for the Students (Yudelson & Ritter, AIED 2015)
    - Small difference in accuracy could mean tangible changes in how students work
    - Larger pLearn → faster mastery → fewer problems and less time

Comparing to the baseline

Accuracies/RMSE of the Majority Class prediction (here, always predicting "correct")

```
$> awk -F"\t" '{N++; C+=$1==1; SSE+= (1-$1)*(1-$1); }END{print C/N"\t"sqrt(SSE/N);}' d/b89_train.txt
```

Accuracy=0.861663 and **RMSE=0.371936**. Majority class accuracy is slightly worse than BKT models we've seen. The majority class RMSE is tangibly worse.

Note! BKT, traditionally, uses first attempt at a problem step. Further work until the student gets the step right is omitted.

- MC Accuracy ≈86% doesn't mean data is severely skewed or the Cognitive Tutor tutor is simple or all students are incredibly smart
  - 20 012 498 first attempts, 17 244 034 are correct = 86.1663% correct
  - 26 798 736 all attempts, 17 244 034 are correct = 64.3464% correct
- Contextual slip and guess BKT (Baker et al., 2010) uses all data.

Let forgetting happen

Default initial parameter values are: pL_0=0.5, 1-pF=1, pT=0.4, 1-pS=0.8, pG=0.2. To set defaults, one can specify a comma separated list via `-0 0.5,1.0,0.4,0.8,0.2`

Let's set a starting version of forgetting to 0.001

```
$> ./trainhmm -s 1.2 -d ~ -m 1 -p 1
-0 0.5,0.999,0.4,0.8,0.2 d/b89_train.txt
m/b89_modelGDpf0.01.txt
p/b89_predictGDpf0.01.txt
```

Π

Known	Unknown
pL_0	1- pL_0

A

	to Known	to Unknown
from Known	**1-pF**	pF
from Unknown	**pT**	1-pT

B

	Correct	Incorrect
Known	**1-pS**	pS
Unknown	**pG**	1-pG

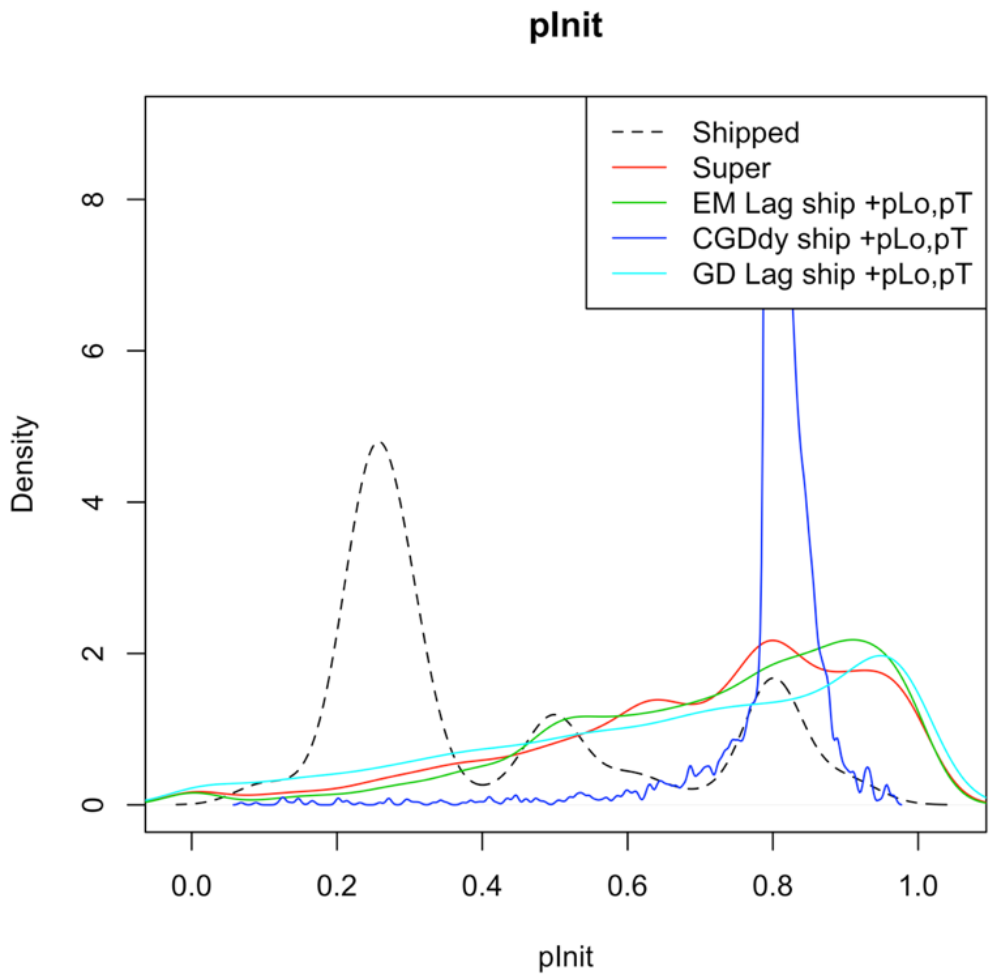
Small decrement in accuracy (rf. to 0.863745)

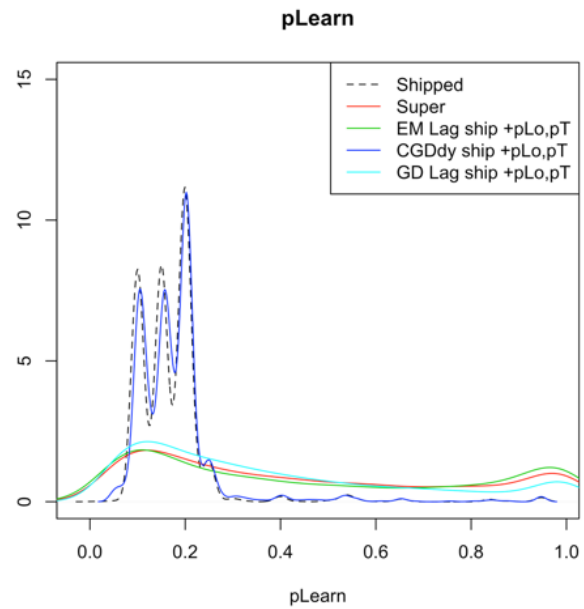
```
trained model LL=7747431.9193728 (5335068.6927902),
AIC=15509615.838746, BIC=15618868.173694, RMSE=0.340602
(0.387471), Acc=0.861669 (0.814925)
timing: overall 101.492081 sec, read 44.915405 sec, fit
30.866999 sec, predict 25.471139 sec
```

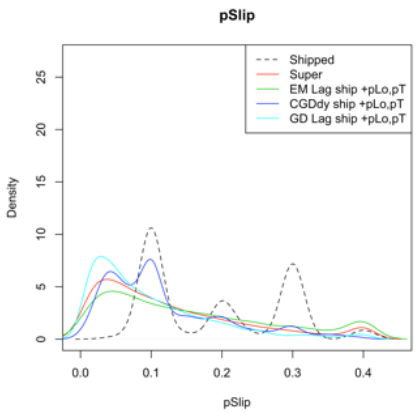
Why didn't we see drastic improvements?

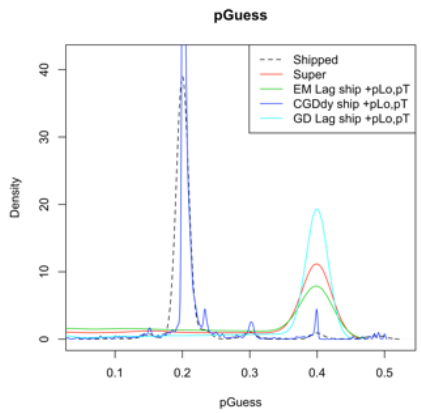
- We are using same set of starting parameters
  - Harder and easier skills *start* in the same place
  - Parameter space has multiple local optima
  - \+ Seeding pL_0 and pT using AFM – *pre-classify* (e.g., via modified `liblinear` in the IEDMS GitHub)
- We are not taking *context* into consideration
  - Learning and forgetting are likely non-uniform
  - Spacing effect: same day, next week, after Thanksgiving
  - \+ Multiplexing (switching) Transition (A) and Emission (B) matrices
- Presence of noise in the data (Falakmasir et al. 2015)

How different model parameters are?









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- Bayesian Knowledge Tracing (BKT)
- BKT Variants
- Fitting BKT at Scale (`hmm-scalable`)
- **Advanced Topics**

Advanced Topics
Prediction

- Updating probability distribution of states (PDS)
  - In tutor, previous observation (correctness) is always known
  - When assessing the model, revealing observation is not fair (e.g., AFM and PFA always *fly blind*)
- Controlling how update of the PDS is done

  `-U r,t` – for known observation **reveal** them for PDS update, for unknown observations – use **transition** matrix only (in Tutor)

  `-U g,g` – for known and unknown observations – **guess** most probable observation from PDO (Model Assessment)
- Probability distribution of observations (PDO) are always computed before the update and without *peeking*

Advanced Topics
Cross-Validation. Overview

- Cross-validation
  - Divide data into *n* pieces (folds)
  - Hide $1/n^{th}$ of the data, train on the rest
  - Predict the hidden data
  - Pick next fold, repeat
  - Combine *n* prediction folds, compute accuracy metric(s)
- Stratification
  - By student – $1/n^{th}$ of the students (whole student-skill sequences)
  - By-item – $1/n^{th}$ of the items (partial student-skill sequences)
  - Un-stratified – $1/n^{th}$ of the rows (partial student-skill sequences)
- Implementation
  - By-student stratification – simple
  - By-item/un-stratified – save $1/n^{th}$ of observations, and mark them as unknown

Advanced Topics
Cross-Validation. Example

- 5-fold student-stratified (default) cross-validation (rf. EM Acc.=0.870607)

```
./trainhmm -s 1.1 -d ~ -p 1 -b 1 -v 5 d/b89_train.bin m/b89_modelEMv.txt
p/b89_predictEMv.txt
...
5-fold cross-validation: LL=6932561.9857819, AIC=13879875.971564,
BIC=13989128.306512, RMSE=0.319513 (0.354041), Acc=0.870297 (0.830288)
timing: overall 206.000000 seconds, read 1.271167, fit 186.711080, predict
17.153612
```

- **10**-fold **item**-stratified cross-validation when predicting **incorrect observation**, and **save** the folds into a **file**

```
./trainhmm -s 1.1 -d ~ -p 1 -b 1 -P 1 -v 10,i,2,folds.txt,o
d/b89_train.bin m/b89_modelEMv2.txt p/b89_predictEMv2.txt
...
10-fold cross-validation: LL=19324495.2940466, AIC=38663742.588093,
BIC=38772994.923041, RMSE=0.321796 (0.357636), Acc=0.871027 (0.831588)
timing: overall 638.000000 seconds, read 1.312280, fit 618.701899,
predict 17.522869
```

- Saved folding can later be reused (read **in**) by using the following switch

```
-v 10,i,2,folds.txt,i
```

Advanced Topics
Parallel Execution

- Parallel code uses Open MP
  - Individual skills are fit in N (for me N=8) dynamically allocated parallel threads
- Checked-in code is in sequential state
  - Sequential code is easier to debug
  - Convert to parallel state by running `./seq2par.sh`
  - Convert back to sequential state `./par2seq.sh`
- To enable parallel execution use `-P 1` switch

```
 ./trainhmm -s 1.1 -d ~ -p 1 -b 1 -P 1 -m 1
d/b89_train.bin m/b89_modelEM.txt p/b89_predictEM.txt
...
timing: overall 33.484420 sec, read 1.232642 sec, fit
14.389842 sec, predict 36.632139 sec
```

Compare to sequential execution

```
timing: overall 139.576931 sec, read 42.205482 sec, fit
60.997095 sec, predict 36.144902 sec
```

Advanced Topics

What if there are just too many skills/students?

- Skill/student internal index variables
  - 4 byte signed integer
  - 2,147,483,647 (2 billion)
  - Default skill/student map is non-sparse
- 2010 Carnegie Learning unfiltered (Big Dataset)
  - 56000 students * 3700 skills → 56000*3700*2*4≅1.54Gb of memory for PSD alone
- `hmm-scalable` with BOOST* library's sparse 2D map
  - Compact representation
  - Currently not in public code

*http://www.boost.org

What I'd Like to add to `hmm-scalable`

- Multiple groupings of sequences
  - Student, Class, School – hierarchical effects
- Conjunctive BKT
  - Address multiple skills coding or other [item context] factors (FAST by González-Brenes et al, NIPS 2013)
- Continuous observations (e.g. time)
  - Math is already worked out
- Speed improvements
  - Optimized computation of forward-backward variables
  - NVIDIA Compute Unified Device Architecture (CUDA)
    - On this laptop: 384 GPU cores vs. 8 CPU cores

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